

Discovery Executive Summary

Project: Hiii · Generated: 9/28/2026, 12:19:33 PM

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Architecture & Design Analysis	—

1. Architecture & Design Analysis

EXECUTIVE SUMMARY

Invoice Ninja is a mature Laravel monolith that already invests in a Service Layer (328 service classes) and a Repository Layer (39 repositories), so the classic "no service tier at all" failure mode is largely avoided — the real risk lives elsewhere. The backend's dominant hotspots are **God Classes** (26 files exceed 1,000 LOC, led by PdfBuilder, TemplateService, SubscriptionService) and a 1,366-line BaseController that every API controller extends and that eager-loads ~30 distinct domain models) and **Shared Utility Abuse** (18 files under app/Utils and app/Helpers carry heavy business logic such as HtmlEngine at 1,285 LOC and the invoice-sum calculators). Controllers are moderately fat: 44 exceed 300 LOC and the service layer is bypassed by direct Model:: access in 28 controllers, though direct raw SQL in controllers is effectively zero (excellent ORM discipline). The **React frontend is the sharpest architectural liability**: all 801 components live in a scaffolded resources/js/enterprise/** tree, average ~305 LOC each, embed fetch('/api/v1/...') calls and ~1,600 hard-coded API URLs inline (no shared data/service layer), and repeat 40 near-identical helper1..40 calculation functions per file. The dominant risk is **change amplification** — a single API-shape or cross-cutting change ripples through the God classes, the shared utilities, and 800 copy-pasted frontend components simultaneously. Layers covered: **Backend** (2,442 PHP files in app/) and **Frontend** (801 React .tsx components); the client portal is a separate .ts/.js bundle (54 files) noted but not the primary focus.

\$1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	182 avg (raw); 44 controllers >300 LOC; max 1,366 (BaseController)	Moderate
H2	Missing Service Layer	Controllers accessing repos/models directly	<10	10–20	>20	28 controllers use direct Model:: access (service layer exists but bypassed)	Moderate
H3	Missing Repository Pattern	Direct DB access points outside repositories	<10	10–20	>20	15 files with DB:: outside repos; widespread Model:: active-record use	Moderate
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0 detected via import scan (resolved through Laravel DI container)	Good
H5	Shared Utility Abuse	Utility/helper files holding business logic	0	1–5	>5	18 files in app/Utils + app/Helpers >300 LOC with business logic	High Risk
H6	Direct SQL in Controllers	ORM compliance %	>90%	60–90%	<60%	100% — 0 DB:: raw queries in any controller	Good
H7	God Classes	Classes >1000 LOC	0	1–3	>3	26 files >1,000 LOC (excl. 3 pure data-array providers)	High Risk
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	BaseController eager-loads ~30 domain models; no bounded contexts	Moderate
H9	Shared Database Coupling	Tables shared across domains	<10%	10–30%	>30%	Single multi-tenant schema (~300 migrations), no per-domain ownership	Moderate
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	305 avg LOC across 801 components; helper1..40 calc logic inline	High

							Risk
F2	Missing Frontend Service/Data Layer	Components w/ inline API calls	<10	10–20	>20	800 components with inline <code>fetch('/api/v1/...')</code> ; ~1,600 hard-coded URLs	High Risk
F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	0 components exceed 400 LOC	Good
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	≤1 — 0 <code>useContext</code> / <code>createContext</code> , self-contained pages	Good
F5	Legacy / Inconsistent Component Patterns	Legacy/inconsistent-pattern components	0	1–10	>10	0 class/deprecated components, but 0 error boundaries across 801 + copy-paste duplication	Moderate

No additional hotspots beyond the standard set were observed.

\$1.4 Actions Required

Hotspot	Action	Rating	Priority
H7 God Classes	Decompose <code>PdfBuilder</code> (2,410), <code>TemplateService</code> (1,828), <code>SubscriptionService</code> , <code>BaseController</code> and 22 other >1,000-LOC files by responsibility (SRP)	High Risk	Critical
F2 Missing Frontend Service/Data Layer	Replace 1,600 inline <code>fetch('/api/v1/...')</code> calls with a shared typed API client + query-hook factory; remove mock fallbacks	High Risk	Critical
H5 Shared Utility Abuse	Extract <code>HtmlEngine</code> , <code>InvoiceSum</code> , <code>GeneratesCounter</code> and 15 other util/helper engines into owned domain services with interfaces	High Risk	High
F1 Business Logic in Components	Extract the <code>helper1..40</code> scoring family and inline filtering into shared hooks/utilities; reduce pages to presentation	High Risk	High
H1 Fat Controllers	Extract <code>bulk()</code> / <code>update()</code> orchestration (incl. the <code>usleep</code> polling loop) from <code>InvoiceController</code> & siblings into Application Services	Moderate	High
H2 Missing Service Layer (bypassed)	Route the 28 controllers' direct <code>Model::</code> queries through the already-injected repositories/services; add a lint guard	Moderate	Medium
H3 Missing Repository Pattern (bypassed)	Move <code>DB::raw</code> reporting/chart SQL and <code>Invoice.php</code> queries behind a reporting repository/read-model	Moderate	Medium
H8 Domain Boundary Violations	Delegate <code>BaseController</code> include maps to per-domain transformers behind an <code>IncludeResolver</code> ; define bounded contexts	Moderate	Medium
H9 Shared Database Coupling	Assign per-context table ownership; expose cross-context reads via internal APIs/DTOs (anti-corruption layer)	Moderate	Medium
F5 Legacy / Inconsistent Patterns	Add a shared <code><ErrorBoundary></code> and a reusable <code>ResourcePage</code> primitive; compose pages instead of cloning	Moderate	Medium

\$1.5 Expected Outcomes

- **Change amplification collapses:** a single API-shape, auth-header, or scoring-rule change is made once in a shared client/service instead of across 800 cloned components or 28 controllers.
- **Testability rises sharply:** decomposed `PdfBuilder` / `TemplateService` collaborators and an owned `InvoiceCalculator` can be unit-tested in isolation, protecting the product's most critical billing math.
- **Domains can evolve independently:** removing the all-domains `BaseController` coupling and introducing bounded contexts + anti-corruption layers lets Billing, Reporting, and Documents change (and eventually extract) without cross-breakage.
- **Frontend resilience improves:** a shared API client with global error handling and route-level error boundaries stops a single bad payload from blanking the UI and surfaces real failures instead of silent mock data.
- **Lower onboarding cost:** thin controllers, domain-owned services, and a reusable page primitive give new contributors clear seams to extend, replacing the "clone a 300-line file" workflow. {"stop_reason":"end_turn","session_id":"1aba09c8-8832-4bf0-9978-483ea4f5b313","total_cost_usd":2.577165,"usage":{"input_tokens":13497,"cache_creation_input_tokens":81577,"cache_read_input_tokens":1575450,"output_tokens":36223,"server_tool_use":{"web_search_requests":0,"web_fetch_requests":0},"service_tier":"standard","cache_creation":{"ephemeral_1h_input_tokens":81577,"ephemeral_5m_input_tokens":0},"inference_geo":"not_available"},"iterations":{"input_tokens":2,"output_tokens":3743,"cache_read_input_tokens":100741,"cache_creation_input_tokens":419,"cache_creation":{"ephemeral_5m_input_tokens":0,"ephemeral_1h_input_tokens":419},"type":"message"}},{"speed":"standard"},"modelUsage":{"claude-haiku-4-5-20251001":{"inputTokens":545,"outputTokens":13,"cacheReadInputTokens":0,"cacheCreationInputTokens":0,"webSearchRequests":0,"costUSD":0.00061,"contextWindow":200000,"maxOutputTokens":32000},"claude-opus-4-8[1m]":{"inputTokens":13497,"outputTokens":36223,"cacheReadInputTokens":1575450,"cacheCreationInputTokens":81577,"webSearchRequests":0,"costUSD":2.576555,"contextWindow":1000000,"maxOutputTokens":16384,"terminal_reason":"completed","fast_mode_state":"off","uuid":"90602b5e-b140-41b5-8417-7fac04641f36"}}